

Aim: To write an Embedded C program for interfacing Ultrasonic Sensor using Arduino Uno and Proteus

objective:

1. To interface an HC-SR04 ultrasonic sensor with an Arduino Uno in the Proteus simulation software.
2. To write code that measures the time it takes for sound waves to bounce back and convert it to centimeters.
3. To display the measured distance readings in real time using a Proteus Virtual Terminal.

program:

```
int trig = 10;
int echo = 9;
long duration;
int cm;
void setup() {
// put your setup code here, to run once:
pinMode(trig, OUTPUT);
pinMode(echo, INPUT);

Serial.begin(9600);
}
void loop() { // Sketch to calculate ultrasonic distance
digitalWrite(trig, LOW);
delayMicroseconds(10);
digitalWrite(trig, HIGH);
delayMicroseconds(10);
digitalWrite(trig, LOW);
delayMicroseconds(10);
duration = pulseIn(echo, HIGH);
cm = (duration/2) * 0.034;
// Print Value on Serial Monitor
Serial.print(" Distance = ");
```

```
Serial.print(cm);
```

```
Serial.println(" cm");
```

```
}
```

Output:

